

CY202 — BIOLOGY AND ENVIRONMENTAL SCIENCE

Unit 1: Cells, Biomolecules & DNA Structure

COMPREHENSIVE STUDY NOTES

For B.Tech Students

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SECTION 1: Characteristics of Living Organisms

Before studying biomolecules, you must understand what defines a living system. Biologists recognise nine cardinal properties that collectively distinguish life from non-living matter.

1.1 The Nine Hallmarks of Life

Property	Core Concept	Key Example / Engineering Analogy
Growth	Increase in cell number (hyperplasia) or cell size (hypertrophy)	Plants: indeterminate growth Animals: determinate growth at maturity
Reproduction	Asexual (one parent, identical offspring) or sexual (genetic recombination from two parents)	Binary fission in bacteria vs. meiosis in humans
Metabolism	Sum of all anabolic (build-up) and catabolic (break-down) reactions	Like a chemical plant running 24/7 — ATP is the energy currency
Cellular Organization	All life is cell-based. Prokaryotes (no nucleus) vs. eukaryotes (membrane-bound nucleus)	Bacteria: simplest; Humans: ~37 trillion specialised cells
Consciousness / Responsiveness	Sensing and responding to stimuli. Phototropism in plants; self-consciousness in humans	A thermostat responds to temp — life responds dynamically and adaptively
Homeostasis	Maintaining stable internal conditions despite external fluctuation	Human body: 37°C regulated by sweating (cooling) and shivering (heating)
Adaptation	Heritable changes that improve survival over generations via natural selection	Desert cacti — thick stems, CAM photosynthesis; Arctic bears — thick fur
Movement	External (locomotion) or internal (cytoplasmic streaming, vesicle transport)	Myosin-actin interaction: nature's nanoscale motor protein
Life Span	Definite period from birth/germination to death	Mayfly: 1 day Bristlecone pine: >5,000 years

 **Professor's Tip**

In engineering terms, a living cell is a self-replicating, self-repairing, energy-harvesting nano-machine. No human technology yet matches its efficiency. The moment you internalise this metaphor, bioenergetics becomes intuitive.

SECTION 2: Carbohydrates

Carbohydrates are the body's primary fuel. From an engineering perspective, think of glucose as the universal energy token — analogous to electrical current in a circuit. Everything runs on it.

2.1 Definition & General Formula

Carbohydrates are polyhydroxy aldehydes or ketones, or compounds that hydrolyse to yield them. Empirical formula: $C_n(H_2O)_n$ (hence the name 'hydrates of carbon'). Ratio C:H:O = 1:2:1.

Why the 1:2:1 ratio matters in exams

This formula is the basis for calculating molecular weight of any simple sugar. Glucose = $C_6H_{12}O_6$, MW = 180 g/mol. Fructose and galactose are structural isomers — same formula, different arrangement. This is why stereochemistry is critical in biochemistry.

2.2 Classification by Degree of Polymerisation

Class	Description & Key Members
Monosaccharides	Single unit, cannot hydrolyse further. Trioses (3C): glyceraldehyde. Pentoses (5C): ribose (RNA), deoxyribose (DNA). Hexoses (6C): glucose, fructose, galactose.
Disaccharides	Two monosaccharides joined by a glycosidic bond. Sucrose (Glc+Fru, alpha-1,2), Lactose (Gal+Glc, beta-1,4), Maltose (Glc+Glc, alpha-1,4).
Oligosaccharides	2-9 units. Critical in cell-cell recognition and immune signalling. ABO blood group antigens are oligosaccharides.
Polysaccharides	Hundreds to thousands of units. Starch (energy, plants), Glycogen (energy, animals), Cellulose (structure, plants), Chitin (structure, fungi/arthropods).

2.3 Monosaccharides in Depth

2.3.1 Classification by Functional Group

Aldoses contain an aldehyde group at C-1. Ketoses contain a ketone group, typically at C-2. Glucose is an aldohexose; fructose is a ketohexose. Galactose is also an aldohexose, differing from glucose only at C-4 (epimer at C-4).

2.3.2 Key Hexoses — The 'Big Three'

Sugar	Type	Primary Role
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D-Glucose	Aldohexose (6C)	Universal metabolic fuel; blood sugar; brain's preferred energy source (~120g/day)
D-Fructose	Ketohexose (6C)	Fruit sugar; absorbed via GLUT5 (not insulin-dependent); metabolised in liver
D-Galactose	Aldohexose (6C)	Component of lactose (milk sugar); converted to glucose in liver via Leloir pathway

2.3.3 Ring Forms (Mutarotation) — Frequently Examined

In aqueous solution, the open-chain form of glucose cyclises when the C-5 hydroxyl attacks the C-1 carbonyl, forming a hemiacetal. This creates a new chiral centre at C-1 (the anomeric carbon), giving two anomers:

- **alpha-D-Glucopyranose:** OH at C-1 is axial (same side as C-6 substituent in chair form). More stable in solid state.
- **beta-D-Glucopyranose:** OH at C-1 is equatorial. Predominates in solution (~64%). This is the form polymerised in cellulose.
- **Mutarotation:** Interconversion between alpha and beta forms via the open-chain intermediate. Equilibrium: ~36% alpha, ~64% beta in water.

⚠ Exam Trap: Pyranose vs. Furanose

Pyranose = 6-membered ring (like pyran). Furanose = 5-membered ring (like furan). Glucose mainly forms pyranose; fructose forms both, but furanose form is more stable in the free state. Ribose (in RNA) forms furanose rings. This is ALWAYS asked in unit tests.

2.4 Glycosidic Bonds — The Connector

A glycosidic bond forms by condensation between the anomeric carbon of one sugar and a hydroxyl group of another, with loss of water. The nature of the bond determines digestibility and function:

Bond Type	Significance
alpha-1,4 glycosidic bond	Found in starch (amylose) and maltose. Digestible by human amylases — helical coiling facilitates compact energy storage.
alpha-1,6 glycosidic bond	Branch points in amylopectin (every 24-30 residues) and glycogen (every 8-12 residues). More branching = faster glucose release.
beta-1,4 glycosidic bond	Found in cellulose and chitin. NOT digestible by humans — no beta-glucosidase. Creates linear, rigid fibres for structural support.
alpha-1,2 glycosidic bond	Found in sucrose. Both anomeric carbons are involved — sucrose is therefore a non-reducing sugar.

2.5 Polysaccharides — Structural vs. Storage

Polysaccharide	Linkage & Structure	Function & Location
Amylose	alpha-1,4 linear	Soluble starch fraction (~20-30% of starch); energy storage in plants
Amylopectin	alpha-1,4 linear + alpha-1,6 branches	Insoluble starch fraction (~70-80%); branched for rapid hydrolysis
Glycogen	alpha-1,4 + alpha-1,6 (more frequent branching)	Animal energy reserve; liver (blood glucose homeostasis), muscle (local fuel)
Cellulose	beta-1,4 linear	Plant cell wall structural support; most abundant organic polymer on Earth
Chitin	beta-1,4 N-acetylglucosamine	Fungal cell walls; arthropod exoskeletons; second-most abundant biopolymer

2.6 Biological Functions of Carbohydrates

- **Primary Energy Source:** Glucose -> Glycolysis -> Pyruvate -> TCA Cycle + Oxidative Phosphorylation -> ~30-32 ATP per glucose. Brain consumes ~120g glucose/day — it cannot use fatty acids under normal conditions.
- **Protein Sparing Effect:** Adequate carbohydrate prevents gluconeogenesis from muscle proteins. This is why zero-carb diets are dangerous — muscle wasting occurs.
- **Structural Role:** Glycosaminoglycans (hyaluronic acid, chondroitin sulphate) form the extracellular matrix (ECM). Hyaluronic acid is used in biomedical engineering scaffolds.
- **Cell Signalling:** Oligosaccharides on glycoproteins and glycolipids mediate: immune recognition, cell adhesion, pathogen binding (viruses bind surface sugars), ABO blood group specificity.
- **Enzymatic Digestion:** Salivary alpha-amylase starts starch hydrolysis in the mouth. Pancreatic amylase continues in the small intestine. Brush-border enzymes (maltase, lactase, sucrase) complete the job to monosaccharides.

Clinical Application: Lactose Intolerance

Approx. 65% of adults worldwide are lactase-deficient. Without lactase, lactose (Gal-beta-1,4-Glc) reaches the colon where bacteria ferment it, producing H2, CO2, and short-chain fatty acids — causing bloating, cramps, and diarrhoea. This is a classic 'enzyme-deficiency leads to substrate accumulation' model — remember this pattern for all inborn errors of metabolism.

SECTION 3: Lipids

Lipids are the most energy-dense biomolecules (9 kcal/g vs. 4 kcal/g for carbs/proteins). They are also the architects of every cell membrane. Understanding lipids is essential for biomedical engineering, pharmaceutical science, and nanotechnology (liposomes).

3.1 Definition & Classification

Lipids are a structurally diverse family of organic molecules united by one physical property: solubility in non-polar solvents (chloroform, ether, benzene) and insolubility in water. They are NOT defined by a single functional group, unlike carbohydrates or proteins.

Lipid Class	Structural Definition & Key Examples
Fatty Acids	Long-chain carboxylic acids (4-28C). Monomeric building blocks of most complex lipids. Example: palmitic acid (C16:0), oleic acid (C18:1).
Triacylglycerols (TAGs)	Glycerol + 3 fatty acid esters. Primary energy storage form in adipocytes. Releases 9 kcal/g on oxidation.
Phospholipids	Glycerol + 2 fatty acids + phosphate + polar head group. Amphipathic. Primary structural component of cell membranes.
Steroids	Four fused hydrocarbon rings (3 six-membered, 1 five-membered). Cholesterol is the master precursor of all steroid hormones, bile acids, and vitamin D.
Waxes	Long-chain fatty acid esterified to a long-chain alcohol. Waterproofing function. Example: beeswax, carnauba wax.
Eicosanoids	20-carbon signalling lipids derived from arachidonic acid. Include prostaglandins, leukotrienes, thromboxanes. Regulate inflammation and immunity.
Lipoproteins	Lipid-protein complexes for aqueous transport of hydrophobic lipids in blood. Classes: HDL, LDL, VLDL, chylomicrons.

3.2 Fatty Acid Structure — The Fundamental Building Block

Structure: Long, non-polar hydrocarbon chain (tail) + polar carboxyl group -COOH (head). Notation: C:D (carbons : double bonds). Example: C18:2 = 18 carbons, 2 double bonds.

Type	Structural Feature	Properties & Examples
Saturated FA	No C=C double bonds. General formula: $C_nH_{(2n+1)}COOH$	Solid at room temp. Tight packing. Palmitic (C16:0), Stearic (C18:0). Found in animal fats.

Monounsaturated FA	ONE C=C double bond (usually cis)	Liquid at room temp. Oleic acid (C18:1, omega-9). Olive oil. Heart-healthy.
Polyunsaturated FA (PUFA)	TWO or more C=C double bonds	Essential FAs (must get from diet). Linoleic (C18:2, omega-6), alpha-Linolenic (C18:3, omega-3). Lowers LDL.
Trans Fatty Acids	Unsaturated but with trans geometry at double bond	Formed by partial hydrogenation. Raise LDL, lower HDL. Associated with CVD risk.

⚠ Cis vs. Trans Geometry — A Critical Distinction

In cis fatty acids, both H atoms are on the SAME side of the double bond, creating a kink/bend in the chain. This prevents tight packing, keeping membranes fluid. In trans fatty acids, H atoms are on OPPOSITE sides — the chain is straight like saturated FA, packs tightly, and raises 'bad' LDL cholesterol. This is the biochemical reason trans fats are banned in many countries.

3.3 Complex Lipid Classes in Detail

3.3.1 Triacylglycerols (Triglycerides)

Structure: Glycerol backbone with three fatty acids attached via ester bonds (-O-CO-). Formed by dehydration (esterification). Hydrolysis by lipases (hormone-sensitive lipase in adipocytes; lipoprotein lipase in capillaries) releases free fatty acids for energy.

- Stored in adipose tissue — nearly anhydrous (no water), hence very compact energy storage.
- Yield ~2.5x more energy per gram than carbohydrates because carbon atoms are in a more reduced (higher hydrogen content) state.

3.3.2 Phospholipids — Membrane Architects

Structure: Glycerol + 2 fatty acids (at C-1 and C-2) + phosphate group (at C-3) + polar head group. The head group determines the specific phospholipid: phosphatidylcholine (PC), phosphatidylethanolamine (PE), phosphatidylserine (PS), etc.

- **Amphipathic nature:** hydrophilic head + hydrophobic tail. This drives spontaneous self-assembly into bilayers in aqueous environments.
- Lipid bilayer: two monolayers with tails facing inward (shielded from water), heads facing the aqueous environment. This is the basis of ALL cell membranes.
- Also form micelles (spherical, one layer) when only one tail is present (e.g., detergent molecules) — basis of soap action.

3.3.3 Cholesterol — The Membrane Modulator

Cholesterol has a rigid steroid ring system + a flexible hydrocarbon tail + one hydroxyl group. It inserts between phospholipids in the membrane with its -OH at the water interface.

- At HIGH temperatures: reduces fluidity (prevents phospholipids from moving too freely).
- At LOW temperatures: increases fluidity (prevents excessive packing and solidification).
- Net effect: BUFFERS membrane fluidity — acts as a 'fluidity thermostat'.
- Precursor of steroid hormones (cortisol, oestrogen, testosterone, progesterone, aldosterone), bile acids, and vitamin D3.

3.4 Lipoprotein Classes — Critical for Clinical Understanding

Lipoprotein	Composition & Size	Function
Chylomicrons	~80-90% TG; largest particle; assembled in intestine	Transport dietary lipids from gut to peripheral tissues
VLDL	~50-60% TG; assembled in liver	Transport endogenous TG from liver to peripheral tissues
IDL	Intermediate; formed from VLDL	Transitional — either taken up by liver or converted to LDL
LDL ('bad')	~45% cholesterol; delivers to cells	Delivers cholesterol to peripheral cells via LDL receptor. High LDL = plaque risk.
HDL ('good')	~50% protein; smallest particle	Reverse cholesterol transport — removes cholesterol FROM peripheral tissues TO liver.

 Professor's Tip

Remember HDL vs LDL: HDL = 'Healthy/Happy', LDL = 'Lousy/Lethal'. HDL removes cholesterol from arteries; LDL deposits it. Statins work by inhibiting HMG-CoA reductase (rate-limiting enzyme of cholesterol synthesis), which upregulates LDL receptors, clearing LDL from blood.

SECTION 4: Proteins

Proteins are the molecular machines of life. Every enzyme, every structural component, every signal transducer — nearly everything that 'does something' in a cell is a protein. Mastery of protein structure directly maps to understanding virtually every disease mechanism and bioengineering application.

4.1 Introduction

Proteins are linear polymers (polypeptides) of amino acids linked by covalent peptide bonds. The human proteome contains an estimated 20,000-25,000 distinct protein-coding genes, but post-translational modifications create hundreds of thousands of distinct functional proteins.

Key protein functions: Enzymes (catalysis), Structural proteins (collagen, keratin, actin), Transport (haemoglobin, albumin), Receptors (cell signalling), Antibodies (immune defence), Hormones (insulin, growth hormone), Contractile (actin-myosin), Storage (ferritin for iron, casein in milk).

4.2 Amino Acids — The Building Blocks

4.2.1 General Structure

All 20 standard amino acids share a common backbone centred on the alpha-carbon (C alpha), which bears four groups:

- **-NH₂ (amino group)**: acts as a base (accepts H⁺)
- **-COOH (carboxyl group)**: acts as an acid (donates H⁺)
- **-H (hydrogen atom)**: non-reactive
- **-R group (side chain)**: varies — determines the amino acid's identity, chemical properties, and role in protein structure

Except glycine (R = -H), all amino acids are chiral (the alpha-carbon is a chiral centre). Biological proteins use exclusively L-amino acids. D-amino acids occur in some bacterial cell walls and antibiotics.

4.2.2 Classification of Amino Acids by R-Group

Category	Members (with single-letter codes)	Key Properties & Exam Relevance
Nonpolar, aliphatic	Gly (G), Ala (A), Val (V), Leu (L), Ile (I), Pro (P), Met (M)	Hydrophobic — buried in protein interior. Pro creates rigid kinks — breaks alpha helices. Met is always the start amino acid in protein synthesis (start codon AUG).
Aromatic	Phe (F), Tyr (Y), Trp (W)	Hydrophobic + aromatic stacking interactions. Tyr and Trp absorb UV at 280nm — basis of protein concentration measurement. Tyr can be phosphorylated (signalling).
Polar, uncharged	Ser (S), Thr (T), Cys (C), Asn (N), Gln (Q)	Can form H-bonds. Cys forms disulphide (-S-S-) bonds — critical for protein stability. Ser and Thr are major phosphorylation sites (kinase signalling).

Positively charged (basic)	Lys (K), Arg (R), His (H)	At physiological pH 7.4, Lys and Arg are +ve. His is unique — pKa ~6, can be protonated or unprotonated at physiological pH, making it ideal for enzymatic catalysis.
Negatively charged (acidic)	Asp (D), Glu (E)	At pH 7.4, both carry negative charge. Aspartate is substrate in transamination reactions. Glutamate is the major excitatory neurotransmitter.

4.2.3 Essential vs. Non-Essential Amino Acids

The 9 Essential Amino Acids (must come from diet)

Histidine, Isoleucine, Leucine, Lysine, Methionine, Phenylalanine, Threonine, Tryptophan, Valine. Mnemonic: 'His Ile Leu Lys Met Phe Thr Trp Val' — or use 'PVT TIM HALL': Phe, Val, Thr, Trp, Ile, Met, His, Arg*, Leu, Lys (*conditionally essential). Additionally: Arginine and Glutamine are conditionally essential during illness, trauma, or rapid growth.

4.3 Peptide Bond Formation

A peptide bond is formed between the alpha-carboxyl group of one amino acid and the alpha-amino group of the next, via dehydration condensation (loss of water). The resulting C-N bond has partial double bond character (resonance) — this makes it planar and rigid.

- Planarity of the peptide bond constrains the polypeptide chain's geometry.
- The chain has directionality: N-terminus (free -NH₂) to C-terminus (free -COOH).
- Protein synthesis on ribosomes proceeds 5'->3' on mRNA, and N->C terminus on the protein.
- A dipeptide has 1 peptide bond; tripeptide has 2; polypeptide has n-1 bonds for n amino acids.

⚠ Exam Alert: Phosphodiester vs. Peptide vs. Glycosidic Bonds

Phosphodiester bonds link nucleotides in DNA/RNA (5'-phosphate to 3'-OH, via phosphodiester linkage). Peptide bonds link amino acids in proteins. Glycosidic bonds link monosaccharides in carbohydrates. Know which bond belongs to which biomolecule — this is a standard one-mark question.

4.4 The Four Levels of Protein Structure

Protein structure is hierarchical. Each level builds on the previous, and each is necessary for proper function. Disruption at any level (denaturation) typically abolishes biological activity.

Structural Level	Details
PRIMARY STRUCTURE	The linear sequence of amino acids, read N->C. Encoded directly by the gene (DNA sequence -> mRNA codon sequence -> amino acid sequence via genetic code). According to Anfinsen's Dogma, primary structure contains ALL information needed for a protein to achieve its correct 3D

	shape. A single amino acid change can cause disease (e.g., Sick Cell Anaemia: Glu->Val at position 6 of beta-globin).
SECONDARY STRUCTURE	Local regular folding patterns stabilised by BACKBONE hydrogen bonds (NH...C=O). Two main types: (1) alpha-Helix: right-handed coil, H-bond forms between residue 'n' and residue 'n+4'. 3.6 residues per turn. Highly stable. (2) beta-Sheet: extended strands connected laterally by H-bonds. Parallel (same direction) or antiparallel (opposite directions — more stable due to more linear H-bonds). Also includes beta-turns and random coils.
TERTIARY STRUCTURE	The overall 3D shape of the entire polypeptide chain. Stabilised by: (a) Hydrophobic interactions (dominant — nonpolar R-groups buried in interior); (b) Disulphide bonds (-S-S-) between Cys residues (only covalent non-backbone bond); (c) Hydrogen bonds between polar R-groups; (d) Ionic bonds (salt bridges) between +ve and -ve charged R-groups; (e) Van der Waals forces (weak, additive). Creates the active site in enzymes and binding pockets in receptors.
QUATERNARY STRUCTURE	Association of TWO OR MORE polypeptide chains (subunits) into a functional complex. Stabilised by the same non-covalent forces as tertiary structure, plus inter-subunit disulphide bonds. Example: Haemoglobin = 2 alpha-chains + 2 beta-chains (tetramer). Each subunit carries a haem group (prosthetic group with Fe ²⁺). Cooperative oxygen binding is a QUATERNARY STRUCTURE property.



Professor's Tip

Think of protein structure like architecture: Primary = the list of bricks used (amino acid sequence); Secondary = local archways/columns (helices/sheets); Tertiary = the complete building's shape (overall fold); Quaternary = a complex of multiple buildings (multi-subunit complex).

4.4.1 Denaturation

Denaturation is the disruption of secondary, tertiary, and quaternary structure (but NOT primary structure — peptide bonds remain intact). Causes: heat, extreme pH, organic solvents (ethanol, acetone), detergents (SDS), heavy metals.

- Reversible denaturation: Some proteins (e.g., ribonuclease A) refold spontaneously when the denaturing agent is removed — demonstrating that primary structure encodes 3D structure (Anfinsen experiment, 1972 Nobel Prize).
- Irreversible denaturation: Requires molecular chaperones (e.g., HSP70, GroEL/GroES chaperonins) to assist refolding in vivo, or is permanently inactivated (e.g., boiling an egg).
- Application: Sterilisation of surgical instruments by autoclaving (121°C, 15 min) works by denaturing bacterial proteins and destroying pathogen viability.

4.5 Classification of Proteins

4.5.1 By Composition

- **Simple Proteins:** Yield only amino acids on hydrolysis. Albumins (water-soluble, coagulate on heating — e.g., egg white), Globulins (soluble in dilute salt — e.g., antibodies, haemoglobin), Scleroproteins/Fibrous proteins (insoluble structural proteins — collagen, keratin, elastin), Histones (basic proteins complexed with DNA).
- **Conjugated Proteins (Holoproteins):** Contain a protein portion (apoprotein) + non-protein prosthetic group. The functional complex = holoprotein. Removal of the prosthetic group yields the apoprotein, which is usually inactive.
- **Derived Proteins:** Products of hydrolysis or denaturation of simple/conjugated proteins (peptones, proteoses, peptides). May retain or lose original function.

4.5.2 Major Classes of Conjugated Proteins

Class	Prosthetic Group	Examples & Functions
Glycoproteins	Carbohydrate (oligosaccharide) chains. N-linked (to Asn) in ER; O-linked (to Ser/Thr) in Golgi.	Immunoglobulins (antibodies), mucins (lubrication), erythropoietin (RBC production). Functions: cell-cell recognition, cell adhesion, immune response.
Lipoproteins	Lipids (TG, cholesterol, phospholipids) non-covalently bound to apolipoproteins	Chylomicrons, VLDL, LDL, HDL. Function: transport hydrophobic lipids in aqueous blood.
Metalloproteins	Metal ion cofactor (Fe2+, Zn2+, Cu, Mn)	Haemoglobin (Fe2+ — O2 transport), Cytochrome c (Fe — electron transport chain), Carbonic anhydrase (Zn2+ — CO2 hydration), Superoxide dismutase (Cu/Zn — antioxidant defence).
Nucleoproteins	Nucleic acids (DNA or RNA)	Histones (basic proteins wrapping DNA in chromatin), Ribosomes (rRNA + ribosomal proteins).
Hemoproteins	Haem group (iron-porphyrin ring)	Haemoglobin, Myoglobin, Cytochromes, Catalase, Peroxidase.
Phosphoproteins	Phosphate group esterified to Ser/Thr/Tyr	Casein (milk protein), phosphorylase b. Also many signalling kinase substrates.

4.5.3 By Shape

- **Fibrous Proteins:** Long, thread-like. Insoluble in water. Primarily structural. Beta-sheets or alpha-helices run parallel to fibre axis. Examples: Collagen (tendons, bone matrix — triple helix), Keratin (hair, nails, claws — alpha-helical coils), Elastin (elastic tissues), Silk fibroin (antiparallel beta-sheet — gives tensile strength).
- **Globular Proteins:** Spherical/compact. Generally water-soluble. Most enzymes, transport proteins, antibodies, and hormones are globular. Hydrophobic residues are buried internally; hydrophilic residues face outward.

SECTION 5: Nucleic Acids & DNA Structure

DNA is the molecule of heredity. RNA is the molecule of gene expression. Together, they constitute the information system of all life. As an engineer, you can think of DNA as a read-only memory (ROM) chip — stable, archival storage — while RNA is the working copy loaded into a processor (the ribosome) to execute instructions.

5.1 Nucleotide — The Monomer of DNA and RNA

A nucleotide consists of three components joined by covalent bonds:

- **Nitrogenous base** (purine or pyrimidine — contains N and C in a ring)
- **Pentose sugar** (deoxyribose in DNA; ribose in RNA)
- **Phosphate group** (one or more -PO₄ groups)

A nucleoside = base + sugar (no phosphate). A nucleotide = nucleoside + phosphate.

5.2 Nitrogenous Bases — Purines and Pyrimidines

Base Class	Members & Structure
PURINES	Adenine (A) and Guanine (G). Double-ring structure: a 6-membered pyrimidine ring fused to a 5-membered imidazole ring. Larger of the two classes.
PYRIMIDINES	Cytosine (C), Thymine (T — DNA only), Uracil (U — RNA only). Single 6-membered ring with N at positions 1 and 3.

Base Pairing — Chargaff's Rules (1950)

In any DNA molecule: [A] = [T] and [G] = [C]. Therefore [A+G] = [C+T] (purines = pyrimidines). A pairs with T via 2 hydrogen bonds. G pairs with C via 3 hydrogen bonds (stronger — GC-rich DNA is more stable, has higher T_m melting temperature). In RNA, U replaces T and pairs with A.

5.3 The Phosphodiester Bond — Backbone of DNA/RNA

Nucleotides are linked by phosphodiester bonds. The phosphate group bridges the 3'-OH of one nucleotide to the 5'-OH of the next, forming a 3'->5' phosphodiester linkage. This creates the sugar-phosphate backbone, which carries the genetic information. The backbone is negatively charged (due to phosphate groups) and hydrophilic, while the bases point inward.

- Bond: 3'-oxygen of sugar₁ — Phosphorus — 5'-oxygen of sugar₂ (two ester bonds, hence 'phospho-DIESTER').
- The backbone runs in the 5'-to-3' direction. Reading direction of DNA = 5'->3'.
- DNA polymerase synthesises new strands only in the 5'->3' direction.

⚠ Phosphodiester vs. Phosphoanhydride Bonds

ATP contains phosphoanhydride bonds (P-O-P linkages between phosphate groups) — these store energy (~30.5 kJ/mol on hydrolysis). Phosphodiester bonds (in DNA/RNA backbone) are not high-energy bonds. Do not confuse these. Hydrolysis of ATP's phosphoanhydride bond drives cellular work; hydrolysis of DNA's phosphodiester bond (by nucleases) destroys the backbone.

5.4 The DNA Double Helix (Watson-Crick Model, 1953)

James Watson and Francis Crick (using X-ray diffraction data from Rosalind Franklin and Maurice Wilkins) proposed the double helix model in 1953. Key structural features:

Structural Feature	Details
Two antiparallel strands	One strand runs 5'->3', the complementary strand runs 3'->5' (antiparallel orientation). Like two parallel roads running in opposite directions.
Right-handed helix (B-form)	The most common form in cells (physiological conditions). Also exists as A-form (dehydrated, RNA-DNA hybrid) and Z-form (left-handed, GC-rich regions under torsional stress).
Base stacking	Bases are stacked ~3.4 Angstrom apart, perpendicular to helix axis. Pi-pi stacking interactions between adjacent aromatic bases contribute significantly to helix stability (more than H-bonds).
Major and minor grooves	Helix winding creates two grooves: Major groove (wider, ~22 Angstrom) — accessible to proteins (transcription factors, restriction enzymes). Minor groove (narrower, ~12 Angstrom) — some proteins/drugs bind here.
Pitch (one complete turn)	34 Angstrom per turn; 10 base pairs per turn; 3.4 Angstrom rise per base pair.
Sugar-phosphate backbone	Deoxyribose + phosphate groups form the structural backbone on the outside of the helix. Bases are on the inside.
Complementary base pairing	A=T (2 H-bonds); G-C (3 H-bonds). This antiparallel complementarity is the basis of DNA replication, transcription, and hybridisation techniques (PCR, Southern blot, etc.).

5.5 DNA vs. RNA — Comparison

Feature	DNA	RNA
Sugar	2'-deoxyribose (lacks -OH at C-2)	Ribose (has -OH at C-2)
Bases	A, T, G, C	A, U, G, C (U replaces T)
Strands	Double-stranded (usually)	Single-stranded (usually)
Location	Nucleus (mainly), mitochondria, chloroplasts	Nucleus, cytoplasm, ribosomes

Stability	Very stable (no 2'-OH; resistant to hydrolysis)	Less stable (2'-OH facilitates hydrolysis under alkaline conditions)
Function	Genetic information storage (ROM)	Gene expression: mRNA (message), tRNA (adapter), rRNA (ribosome component)
Helix form	Mainly B-form double helix	A-form; also hairpins, pseudoknots in secondary structure



Professor's Tip

Why is DNA more stable than RNA? The 2'-OH group in RNA can attack the adjacent phosphodiester bond, causing self-hydrolysis (especially at high pH). DNA lacks this 2'-OH, making it inherently more chemically stable — ideal for long-term genetic information storage. This is why evolution selected DNA over RNA as the primary storage molecule.

SECTION 6: Quick Reference — Key Numbers & Facts

4 kcal/g Carbohydrates Energy density per gram	9 kcal/g Lipids Energy density per gram	4 kcal/g Proteins Energy density per gram	~120 g/day Brain Glucose Obligate daily consumption
20 Amino Acids Standard proteinogenic	9 Essential AAs Must come from diet	20,000+ Human Proteins Estimated in the proteome	~65% Lactose Intolerance Global adult prevalence

6.1 Bond Types — Quick Comparison Table

Bond	Found In	Formed By
Peptide bond	Proteins (between amino acids)	Condensation: alpha-COOH + alpha-NH2 -> -CO-NH- + H2O
Glycosidic bond	Carbohydrates (between sugars)	Condensation between anomeric C and -OH of another sugar
Phosphodiester bond	DNA/RNA (between nucleotides)	3'-OH + phosphate + 5'-OH -> 3'-O-P-O-5'
Ester bond	Lipids (glycerol + fatty acid)	Condensation: -COOH + -OH -> -COO- + H2O
Disulphide bond	Proteins (Cys-Cys)	Oxidation of two -SH groups -> -S-S-
Hydrogen bond	DNA (A=T, G=C), Protein 2deg structure	Electrostatic attraction between N-H or O-H donors and N or O acceptors

6.2 Must-Know Mnemonics & Memory Aids

Mnemonics for Exam Success 9 Essential Amino Acids: 'PVT TIM HALL' = Phe, Val, Thr, Trp, Ile, Met, His, Arg*, Leu, Lys Chargaff's Rule: 'Pure As Gold' = Purines pair with pyrimidines; A=T (2H-bonds), G=C (3H-bonds) Protein Structure Hierarchy: 'Please Stop The Questioning' = Primary, Secondary, Tertiary, Quaternary
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Lipid Energy: 'Fats are FAT with energy' = 9 kcal/g (more than 2x carbs/proteins at 4 kcal/g)
alpha vs. beta linkage: alpha-1,4 = Digestible (starch/glycogen); beta-1,4 = NOT digestible (cellulose) by humans
HDL vs. LDL: HDL = Healthy/takes lipids Home to liver; LDL = Lethal/Leaves lipids in arteries

6.3 Common Exam Questions & Model Answers

Typical Exam Question	Key Points for Full Marks
Why can humans digest starch but not cellulose?	Both are glucose polymers but differ in glycosidic linkage: starch has alpha-1,4 bonds (digestible by human amylase); cellulose has beta-1,4 bonds (no human beta-glucosidase exists).
Distinguish between primary and tertiary protein structure.	Primary = amino acid sequence (covalent peptide bonds); Tertiary = 3D fold of entire chain (non-covalent forces + disulphide bonds). Primary encodes tertiary (Anfinsen's dogma).
Why are trans fats harmful?	Trans double bonds make FA chains straight (like saturated FA), raising LDL and lowering HDL cholesterol, increasing cardiovascular disease risk.
What is mutarotation?	Interconversion of alpha and beta anomers of a monosaccharide in solution via the open-chain form. Glucose: equilibrates to ~36% alpha, ~64% beta-D-glucopyranose.
Explain why DNA is more stable than RNA.	DNA lacks the 2'-OH group (has H at C-2 instead). The 2'-OH in RNA can attack the adjacent phosphodiester bond, causing spontaneous hydrolysis. Hence DNA is a more stable genetic storage molecule.
What is Anfinsen's dogma?	Primary structure (amino acid sequence) contains all information necessary to specify the native 3D conformation of a protein. Demonstrated by denaturation/renaturation of ribonuclease A (Nobel Prize 1972).

End of Study Notes — CY202 Unit 1

Revise these notes 3x: once immediately, once after 24 hours, once after one week (spaced repetition).
Good luck!